WHY RACING WORKS

EMBI – Systems Biology
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HEADS UP!

By exploring the demo of RaCInG you have seen how it could potentially be used to infer cell-cell interactions. However, RaCInG is advertised as a method with strong mathematical justification. In this theoretical assignment we try explore why RaCInG is expected to work and be consistent on a vast variety of input data. The exercises in this document are based on the following mathematical paper. You are encouraged to keep this paper open.

Prerequisites

These theoretical exercises will allow you to explore the mathematical backbone of RaCInG. However, this does mean that some mathematical maturity is required. Specifically, we assume the following knowledge from the field of probability theory:

- You should be comfortable with discrete random variables.
- You should be able to compute the expected value of random variables.
- You should be able to write random variables in terms of (sums of) indicator random variables.
- You should be comfortable with conditional probabilities.

1 Inhomogeneous Random Digraphs

Many of the kernel computations in RaCInG are based on *inhomogeneous random digraphs*. Their mathematical definition is given Section 2.1 of the mathematical paper. At its core, this model generates a directed network with n vertices. Each of these vertices v is given a random type T_v from a discrete distribution. Then, conditional on the vertex types, an arc is drawn from vertex v to vertex w with probability $\kappa(T_v, T_w)/n$ if $v \neq w$. In this part, we will use the theory of inhomogeneous random digraphs to explain where the kernel computations come from in RaCInG. As our main example, we will focus on wedges. These are sets of three vertices (u, v, w) such that the arc (u, v) is part of the graph and the arc (v, w) is part of the graph.

Problem 1 (Counting Wedges)**.** *Denote by W_n the number of wedges in an inhomogeneous random digraph, and by I_{vw} the indicator that arc (v, w) is part of the graph. Write W_n as a sum of indicators.*

Problem 2 (Indicator Expectation)**.** *Write $\mathbb{P}(T_v = x) = q_x$, and assume that $v \neq w$. Show that*

$$\mathbb{E}[I_{vw}] = \sum_x \sum_y \frac{\kappa(x,y) q_x q_y}{n}.$$

Problem 3 (Double Indicator Expectation)**.** *Assume that u, v and w are three distinct vertices. Show that*

$$\mathbb{E}[I_{uv} I_{vw}] = \sum_x \sum_y \sum_z \frac{\kappa(x,y)\kappa(y,z) q_x q_y q_z}{n^2}.$$

Problem 4 (Wedge Expectation)**.** *Use the previous problems to show that*

$$\mathbb{E}[W_n/n] \to \sum_x \sum_y \sum_z \kappa(x,y)\kappa(y,z) q_x q_y q_z.$$

Problem 5 (Specific Wedges)**.** *Denote by $W_n^{(x,y,z)}$ the number of wedges that start at a vertex with type x, then move to a vertex with type y, and end at a vertex with type z. Show that $\mathbb{E}[W_n^{(x,y,z)}/n] \to \kappa(x,y)\kappa(y,z)q_x q_y q_z$.*

Note that the limit in Problem 5 agrees with the description used in the RaCInG paper to compute wedges with the exception of the factor $q_x q_y q_z$. It turns out that this factor is not needed if you consider normalized kernel values, since this factor appears both in the wedge limit for the non-uniform run of the model as well as the uniform run. When you divide the two, this factor disappears.

Problem 6 (Same Factor)**.** *Explain why the factor $q_x q_y q_z$ will appear in both runs (uniform and non-uniform) of the model.*